Data Collection, Bias and Ethics

Student work-along checklist

Name: Date started:

How to use this checklist

Keep this sheet beside you as you work through the online unit. Tick each box when you have done it.

1. Write first. Answer each question in your own words (online or on paper) **before** you reveal the model answer.

2. Compare. Read the model answer and look for what is the same and what is different.

3. Annotate. Add one thing you missed or would change. A different answer is not always a wrong answer.

Before you start: read the Introduction and learning objectives, the Recap of Unit 3, and The Research Question and Problem Statement.

The whole unit takes about **3 hours**. You do not have to do it in one sitting. Use the section headings to find your place again.

Keep in view: the agreed research question and problem statement

Research question: "Can health and lifestyle data be used to estimate an individual's Health Score?"

Problem statement: "Investigate whether health and lifestyle data can be used to estimate an individual's Health Score. The investigation will use the available health and lifestyle data to determine whether useful estimates can be produced that support future health research and health and wellbeing initiatives."

☐ I have compared my own research question and problem statement from Unit 3 with these agreed versions.

You will need: health_data.csv and the Orange **.ows** workflow you saved in Unit 3.

1 Data: What Is Needed and How to Obtain It

☐ I can work out which variables my research question needs, and I know the best data I can **realistically** get is not always the data I would ideally choose.

☐ I can explain the difference between **primary** and **secondary** data.

Activity 4.1 question ☐ 4.1

2 Data Collection

☐ I can give one benefit and one limitation of each method: surveys and questionnaires, interviews, observation, experiments, sensors and wearable devices, web scraping, APIs, existing databases and open datasets, **derived variables** and **synthetic data generation**.

☐ I can explain what **synthetic data** is, and why its conclusions must be checked against real data before they are trusted.

3 The Health Dataset

Activity 4.2: how might each variable have been collected?

Open your Unit 3 workflow. If you no longer have it, load health_data.csv with a CSV File Import widget (right-click the first row indicator and make sure 'Header' is selected) and connect it to a Data Table widget.

☐ Open the Data Table and find **six** variables that may have been collected using different methods, and write down their names.

☐ Save the workflow. You will keep adding to it in this unit.

Activity 4.2 questions ☐ 4.2(a) ☐ 4.2(b) ☐ 4.2(c) ☐ 4.2(d) ☐ Extension

Data cards, provenance and merging

☐ I can say what a **data card** covers, and explain what a dataset's **provenance** is.

☐ Before **merging** datasets, I check for matching column names, units and category labels (for example “Y” and “YES”).

☐ I have read **Where Our Dataset Really Comes From** and **Why Now?** I can explain why a synthetic dataset was chosen, and why issues were added on purpose.

4 Legal Considerations for Data

☐ I know how to find the licence for our dataset (**CC BY-SA 4.0**): we must credit the creator, and share any adapted version of the dataset under the same licence.

☐ I can explain what a **data controller** is and what they must do: have a lawful reason, keep data secure, use it only as agreed and not keep it too long. (For a school project, this is usually the school.)

☐ I can say what potential participants must be told before they **consent**, who else may need to give permission (for example a school or parent), and that participants can **withdraw** at any time.

Collecting data about real people for your own project? Talk to your school’s Data Protection Officer or safeguarding lead first.

5 Representative Data

☐ I can explain the difference between a **population** and a **sample**, and why a sample must be representative.

☐ I can describe **stratified sampling** and **quota sampling**.

☐ I can explain how **convenience sampling** and **voluntary response bias** can skew a sample, and why checking one variable is not enough.

Activity 4.3: data distributions

☐ Add a **Distributions** widget (Visualize), connect it to CSV File Import, and view Gender, Country, Age and Gym_Member. Widen the window to see the full x-axis.

Activity 4.3 questions ☐ 4.3(a) ☐ 4.3(b) ☐ 4.3(c) ☐ 4.3(d) ☐ 4.3(e)

*Note any group that looks under- or over-represented. Add it to the **Data issues log** after Activity 4.7.*

6 Bias in Data

☐ I can describe the six types of bias: **sampling, self-report (response), non-response, voluntary response, measurement** and **historical**.

☐ I can explain the difference between **non-response** bias (people were invited but did not reply) and **voluntary response** bias (people chose to put themselves forward).

☐ I can explain why collecting **more** data does not automatically remove bias.

☐ When bias cannot be removed, I know to **document** it, avoid firm conclusions about groups the data covers poorly, and **communicate** the limitations.

Activity 4.4 questions ☐ 4.4(a) ☐ 4.4(b)

7 Ethics in Data Collection

☐ I can explain the six principles: **consent, privacy, fairness, transparency, accountability** and **data minimisation**.

☐ I can give an ethical issue that is not about personal data, and one argument for and against **web scraping**.

☐ I can explain why health data is **special category** data, and the difference between **anonymised** and **pseudonymised** data.

Activity 4.5: Person_IDs

☐ In the Data Table, check Person_ID and look for any names or details that identify someone.

Activity 4.5 questions ☐ 4.5(a) ☐ 4.5(b)

Data quality: duplicate records

Activity 4.6: examining rows

☐ Sort Person_ID into ascending order. Find **P00019** and compare it with the row below.

Activity 4.6 questions ☐ 4.6(a) ☐ 4.6(b) ☐ 4.6(c)

Activity 4.7: duplicate detection

Orange steps ☐ Group By added ☐ Count only ☐ New Data Table ☐ Sorted by Count

*Group By is in the **Transform** section. Group by Person_ID, choose Count for Person_ID and untick any other aggregations. Sort Count so the highest values are at the top.*

Activity 4.7 question ☐ 4.7

☐ I can explain why the duplicates make the dataset less representative (more on this in Units 5 and 6).

Data issues log

Record what you found. You will need these notes when you start preparing the data.

Issue	What I found (use the starter if it helps)
1. Representation Activity 4.3. Which groups look under- or over-represented?	*The group(s) …*
2. Category labels Activity 4.3(c). Are any labels recorded inconsistently?	*In the Country column …*
3. Possible bias Activity 4.4. Which columns might be affected, and how?	*The … column may be affected by … bias because …*
4. Duplicates Activities 4.6 and 4.7. How many, and what could they affect?	*I found … duplicate Person_IDs. This could …*

8 Data Storage

☐ I can sort storage decisions into three groups: **legal requirements** (security, retention, purpose), **professional practice** (backups, open format, keeping raw data separate, documentation) and **personal choices** (where to store it, which tools, file naming, version tracking).

☐ Before uploading data, I check where a cloud service or AI tool will store it and what it may do with it.

☐ My original health_data.csv is untouched. I work on a separate, version-numbered copy.

9 Unit 4 Review

☐ I have read the **Summary of the Unit** and answered the seven **Reflection Questions** (good to discuss with a partner or mentor).

Optional extensions ☐ 1: special category data law ☐ 2: the scraping tribunal

How confident am I?

Tick each one you are confident about. Leave a box empty if you are not sure yet, and ask your teacher or mentor about it.

☐ I can describe common methods of collecting data
☐ I can explain population and sample
☐ I can explain why a sample must be representative
☐ I can explain licences and legal duties for data
☐ I can identify common sources of bias
☐ I can explain the key ethical principles
☐ I can explain why health data is sensitive
☐ I can check a dataset for bias and ethical issues
☐ I can explain how data should be stored

Ready for Unit 5?

☐ My original **health_data.csv** is still untouched in my course folder.
☐ My updated Orange **.ows** workflow (with the Distributions and Group By widgets) is saved in the same folder.
☐ I have filled in the **Data issues log** (Section 7).
☐ I have noted anything I am still unsure about to ask my teacher or mentor.